

SQL JOIN Notes

student

roll int primary key,auto_increment
name varchar(30)
mail varchar(30)
department_id (int)
marks int

department

department_id int primary key ,auto_increment
name varchar(30)

What is JOIN in SQL?

A JOIN is used in SQL to combine rows from two or more tables based on a related column between them.

It helps to retrieve meaningful data that is spread across multiple tables.

Why JOIN is used?

In relational databases, data is stored in multiple tables to reduce redundancy.

Example:

Student table stores student details

Department table stores department details

To get:

Student name + Department name

JOIN works when:

Tables have a common column

Values in that column are related

Example:

Student.department_id

Department.id

Syntax:

```
SELECT columns
FROM table1
JOIN table2
ON table1.common_column = table2.common_column;
```

Update student marks for a specific roll number //101=>10

Update department of a student

```
>update student set department_id=20 where roll=100;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

Get students from a specific department //any department

```
select * from student where department_id=10;
+----+-----+-----+-----+-----+
| roll | name   | email      | Marks | department_id |
+----+-----+-----+-----+-----+
| 100 | vishakha | v@gmail.com | 89 | 10 |
+----+-----+-----+-----+-----+
```

Get student by email

```
>select * from student where email='v@gmail.com';
+----+-----+-----+-----+-----+
| roll | name   | email      | Marks | department_id |
+----+-----+-----+-----+-----+
| 100 | vishakha | v@gmail.com | 89 | 20 |
+----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

Delete student by roll number

```
delete from student where roll=101;
Query OK, 1 row affected (0.01 sec)
```

Get student details with department name

```
select s.roll,s.name,s.email,s.marks ,d.name from student s join department d on s.department_id
=d.department_id;
```

```
+-----+-----+-----+-----+
| roll | name   | email   | marks | name   |
+-----+-----+-----+-----+
| 100 | vishakha | v@gmail.com | 89 | IT      |
| 102 | Nisha   | n@gmail.com | 90 | Electrical |
+-----+-----+-----+-----+
```

2 rows in set (0.00 sec)
